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|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------|-------------|
| Programme      | B. Sc. Mathematics Honours                                                                                                                                                                                                           |                           |                    |             |
| Course Code    | MAT6EJ301(1)                                                                                                                                                                                                                         |                           |                    |             |
| Course Title   | <b>NUMERICAL ANALYSIS</b>                                                                                                                                                                                                            |                           |                    |             |
| Type of Course | <b>Elective (Specialisation- Mathematical Computing)</b>                                                                                                                                                                             |                           |                    |             |
| Semester       | VI                                                                                                                                                                                                                                   |                           |                    |             |
| Academic Level | 300- 399                                                                                                                                                                                                                             |                           |                    |             |
| Course Details | Credit                                                                                                                                                                                                                               | Lecture/Tutorial per week | Practical per week | Total Hours |
|                | 4                                                                                                                                                                                                                                    | 4                         | -                  | 60          |
| Pre-requisites | 1. Real analysis<br>2. Linear algebra<br>3. Basics of Python Programming                                                                                                                                                             |                           |                    |             |
| Course Summary | This course familiarises students with the fundamental numerical analysis. Moreover, the course facilitates students to apply results from real analysis and linear algebra to perform quantitative analysis of numerical solutions. |                           |                    |             |

| CO                                                                                                                                                                                             | CO Statement                                                                                                                                                               | Cognitive Level* | Knowledge Category# | Evaluation Tools used                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|--------------------------------------------------------|
| CO1                                                                                                                                                                                            | Understand and apply the Bisection Method, Iteration Method, Newton-Raphson Method, and Secant Method to solve algebraic and transcendental equations numerically.         | Ap               | P                   | Internal Exam/Assignment /Seminar/ Viva / End Sem Exam |
| CO2                                                                                                                                                                                            | Implement interpolation methods such as Newton's formulae, Lagrange's interpolation formula, and divided differences to approximate functions from discrete data.          | Ap               | P                   | Internal Exam/Assignment /Seminar/ Viva / End Sem Exam |
| CO3                                                                                                                                                                                            | Implement numerical methods such as Euler's method, Modified Euler's Method, Runge-Kutta method, and Adams-Moulton Method to solve ordinary differential equations (ODEs). | Ap               | P                   | Internal Exam/Assignment /Seminar/ Viva / End Sem Exam |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M) |                                                                                                                                                                            |                  |                     |                                                        |

**Detailed Syllabus:**

|                  |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                         |
|------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Text Book</b> |                                                                               | <p>[1]. S. S. Sastry, <b>Introductory Methods of Numerical Analysis (5/e)</b>, PHI Learning (2012)</p> <p>[2]. Dimitrios Mitsotakis: <b>Computational Mathematics: An Introduction to Numerical Analysis and Scientific Computing with Python</b>, CRC Press (2023), ISBN 978-1-032-26240-6.</p> <p>[3]. Jupyter Notebooks of [2] available at:</p> <p><a href="https://github.com/dmitsot/computational_mathematics">https://github.com/dmitsot/computational_mathematics</a></p> |                         |
| <b>Module</b>    | <b>Unit</b>                                                                   | <b>Content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Hrs<br/>(48 +12)</b> |
| <b>I</b>         | <b>Numerical Solutions of Algebraic and Transcendental equations (Text 1)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>12</b>               |
|                  | 1                                                                             | 2.1 Introduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |
|                  | 2                                                                             | 2.2 Bisection Method                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                         |
|                  | 3                                                                             | 2.4 Iteration Method (Derivation of Condition for Convergence and Acceleration of Convergence are optional)                                                                                                                                                                                                                                                                                                                                                                        |                         |
|                  | 4                                                                             | 2.5 Newton- Raphson Method (Generalized Newton's Method is optional)                                                                                                                                                                                                                                                                                                                                                                                                               |                         |
|                  | 5                                                                             | 2.7 Secant Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                         |
| <b>II</b>        | <b>Interpolation (Text 1)</b>                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>12</b>               |
|                  | 6                                                                             | 3.1 Introduction, 3.3.1 Forward differences, 3.3.2 Backward differences                                                                                                                                                                                                                                                                                                                                                                                                            |                         |
|                  | 7                                                                             | 3.6 Newton's formulae for interpolation (up to and including Example 3.5)                                                                                                                                                                                                                                                                                                                                                                                                          |                         |
|                  | 8                                                                             | 3.6 Newton's formulae for interpolation (From Example 3.6)                                                                                                                                                                                                                                                                                                                                                                                                                         |                         |
|                  | 9                                                                             | 3.9.1 Langrange's interpolation formula                                                                                                                                                                                                                                                                                                                                                                                                                                            |                         |
|                  | 10                                                                            | 3.10 Divided differences and their properties                                                                                                                                                                                                                                                                                                                                                                                                                                      |                         |
|                  | 11                                                                            | 3.10.1 Newton's General interpolation formula                                                                                                                                                                                                                                                                                                                                                                                                                                      |                         |
| <b>III</b>       | <b>Numerical Differentiation and Integration (Text 1)</b>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>12</b>               |
|                  | 12                                                                            | 6.1 Introduction, 6.2 Numerical Differentiation (6.2.1, 6.2.2 and 6.2.3 are optional)                                                                                                                                                                                                                                                                                                                                                                                              |                         |
|                  | 13                                                                            | 6.4.1 Trapezoidal Rule                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |
|                  | 14                                                                            | 6.4.2 Simpson's 1/3-Rule                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |
|                  | 15                                                                            | 6.4.3 Simpson's 3/8 Rule                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |
|                  | 16                                                                            | 6.10 Numerical Double Integration                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                         |
| <b>IV</b>        | <b>Numerical Solutions of Ordinary Differential Equation (Text 1)</b>         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>12</b>               |
|                  | 17                                                                            | 8.1 Introduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |
|                  | 18                                                                            | 8.2 Solution by Taylor's series,                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |
|                  | 19                                                                            | 8.4 Euler's method (8.4.1 is optional)                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |
|                  | 20                                                                            | 8.4.2 Modified Euler's Method                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                         |
|                  | 21                                                                            | 8.5 Runge-Kutta method                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                         |
|                  | 22                                                                            | 8.6.1 Adams-Moulton Method                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                         |
| <b>V</b>         | <b>Numerical Algorithms and Lab Practicals</b>                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>12</b>               |

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|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Jupyter Lab and Notebooks. Google Colab. Instructions in [6] and [7]. Quick review of Python Programming. Ch 1 Notebook from [3].                                                                      |
| 2  | Continue Quick Review of Python. Notebook [9]. Numpy and Scipy review from [7]. Ch 2 Notebook from [3].                                                                                                |
| 3  | Bisection Method. Algorithm and Program.<br>Jupyter Notebook: Ch 5 of [3]. Refer also 5.1 of [2].<br>Optional: Program to compute speed of convergence.<br>Optional: False Position variant from [12]. |
| 4  | Fixed Point Method (Iteration Method). Algorithm and Program.<br>Notebook: Ch 5 of [3]. Reference: 5.2 of [2].                                                                                         |
| 5  | Newton-Raphson Method. Algorithm and Program.<br>Notebook: Ch 5 of [3]. Reference: 5.3 of [2].                                                                                                         |
| 6  | Secant Method. Algorithm and Program.<br>Notebook: Ch 5 of [3]. Reference: 5.4 of [2].                                                                                                                 |
| 7  | Fast computation using SciPy.Optimize.<br>Notebook: Ch 5 of [3]. Reference: 5.6 of [2].                                                                                                                |
| 8. | Lagrange Interpolation.<br>Notebook: Ch 6 of [3]. Reference: 6.1 of [2].                                                                                                                               |
| 9  | Newton's method for Interpolation using Divided Differences.<br>Notebook: Ch 6 of [3]. Reference: 6.2 of [2].                                                                                          |
| 10 | Using SciPy.Interpolate Module. Lagrange Interpolation Only.<br>Notebook: Ch 6 of [3]. Reference: 6.6 of [2].                                                                                          |
| 11 | Numerical Differentiation. Forward and Backward Differences. First Order and Second Order Derivative Approximations.<br>Notebook: Ch 8 of [3]. Reference: 8.1 of [2].                                  |
| 12 | Numerical Integration. Midpoint Rule. Composite Trapezoidal Rule. Composite Simpson's Rule.<br>Notebook: Ch 7 of [3]. Reference: 7.1. of [2].                                                          |
| 13 | The Module scipy.integrate.<br>Trapezoidal, Simpson.<br>Reference: 7.4 of [2]. Notebook: Ch 7 of [3].                                                                                                  |
| 14 | Euler's Method. Improved Euler's Method. Reference: 8.2 of [2].<br>Notebook: Ch 8 of [3].                                                                                                              |

References:

1. F.B. Hildebrand: Introduction to Numerical Analysis, TMH.
2. J.B. Scarborough: Numerical Mathematical Analysis, Oxford and IBH
3. Joakim Sundnes, Introduction to Scientific Programming with Python. Springer (2020). ISBN 978-3-030-50355-0. Open Access at: <https://link.springer.com/book/10.1007/978-3-030-50356-7>
4. Sven Linge and Hans Petter Langtangen, Programming for Computations -- Python. A Gentle Introduction to Numerical Simulations With Python. Springer (2018). ISBN 978-3-319-81282-3. Open Access at: <https://link.springer.com/book/10.1007/978-3-319-32428-9>

**Note: 1) Optional topics are exempted for end semester examination.**

**2) 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

**3) Module V is algorithms and lab computations. Algorithms for each numerical method can be taught along with the Python code in lab sessions. The second text [2] stresses computation from the beginning and is a lab reference. The Jupyter Notebooks [3] intended for live lab lessons.**

**Mapping of COs with PSOs and POs:**

|      | PSO1 | PSO2 | PSO3 | PSO4 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|------|------|------|------|------|-----|-----|-----|-----|-----|-----|-----|
| CO 1 | 3    | 3    | 2    | 1    | 3   | 1   | 3   | 3   | 3   | 0   | 2   |
| CO 2 | 2    | 3    | 3    | 2    | 3   | 1   | 3   | 3   | 3   | 0   | 2   |
| CO 3 | 3    | 3    | 3    | 2    | 3   | 1   | 3   | 3   | 3   | 0   | 2   |

**Correlation Levels:**

| Level | Correlation        |
|-------|--------------------|
| -     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

**Mapping of COs to Assessment Rubrics:**

|      | Internal Exam | Assignment | Seminar | Viva | End Semester Examinations |
|------|---------------|------------|---------|------|---------------------------|
| CO 1 | ✓             | ✓          | ✓       | ✓    | ✓                         |
| CO 2 | ✓             | ✓          | ✓       | ✓    | ✓                         |
| CO 3 | ✓             | ✓          | ✓       | ✓    | ✓                         |